Final Technology Stack

Vulnerability Detection Using Edge AI (Desktop Application)

1. Overview

This document outlines the complete technology stack for an offline desktop-based vulnerability detection system using Edge Artificial Intelligence and a fine-tuned Qwen model.

2. Core Development

- Programming Language: Python 3.10+
- Execution Mode: Fully offline (Edge AI)

3. Desktop Application Framework

- PySide6 (Qt Framework)
- Cross-platform native desktop application

4. Edge AI / Large Language Model

- Qwen (Qwen2 / Qwen-Coder) large language model
- Fine-tuned for vulnerability detection and secure code review

5. Model Fine-Tuning Stack

- Hugging Face Transformers
- PEFT techniques (LoRA / QLoRA)
- PyTorch for training and optimization

6. Offline Inference Runtime

- Ollama or llama.cpp
- Local inference without cloud dependency

7. Static Security Analysis

- Semgrep (multi-language security analysis)
- Bandit (Python-specific analysis)

8. Code Processing

- Tree-sitter for AST-based parsing

9. Reporting & Visualization

- PDF security report generation
- Structured JSON result storage

10. Security Standards

- OWASP Top 10
- CWE (Common Weakness Enumeration)

11. Operating System Support

- Windows
- Linux